

Pankaja Isuranga Balasooriya

Electronic and Telecommunication Engineering Undergraduate, University of Moratuwa

📧 PankajaBalasooriya | 🌐 pankajabalasooriya | 🌐 www.pankajabalasooriya.me

✉️ pankajabalasooriya@gmail.com | 📞 +94.70.138.0990 📠 +61.433.86.0990

SUMMARY

Engineering undergraduate with hands-on experience in embedded systems design, real-time firmware development, and full-stack IoT applications. Skilled in embedded firmware, custom electronics design, and application development, with a track record of building end-to-end hardware-software systems.

EXPERIENCE

Monash University

Clayton VIC, Australia

Human Computer Interaction Research Internship at **Exertion Games Lab**

Dec 2025 - Present

- Designed embedded systems (STM32, custom PCBs), developed firmware and full-stack software, conducted user studies, and co-authored papers for ACM UIST and ACM DIS 2026.

University of Moratuwa

Moratuwa, Sri Lanka

Undergraduate Teaching Assistant: EN2533 - Robot Design and Competition

Aug 2025 - Nov 2025

- Competition task design, mentorship, conducting lab sessions, grading

Sri Lanka Insurance Corporation Limited

Colombo, Sri Lanka

Software Development Intern

Aug 2022 - Oct 2022

- Enterprise Web Development, Database Management, Software Testing

EDUCATION

University of Moratuwa

Moratuwa, Sri Lanka

B.Sc. Eng (Hons) in Electronic & Telecommunication Engineering; CGPA: 3.73/4.00

Feb 2023 - Present

- Robotics, Computer Vision, Electronic Control Systems Design, Circuits and Systems Design, Linear Algebra, Signals and Systems

Bandaranayake College

Gampaha, Sri Lanka

G.C.E Advanced Level - Physical Sciences; Z-Score: 2.5230

May 2019 - Sep 2022

- Combined Mathematics, Physics, Chemistry

G.C.E Ordinary Level; Grade: 9As

May 2014 - Dec 2018

- Mathematics, Science, English Literature, Business Studies, ICT

ESOFT Metro Campus

Gampaha, Sri Lanka

Pearson Assured Diploma in Software Engineering

Dec 2018 - Oct 2019

- OOP, Database, Software Engineering Principles

Pearson Assured Diploma in Information Technology; Batch Top grade honors

Dec 2018 - Aug 2019

PUBLICATIONS

Balasooriya, P., Pathirana, R.O., Warushavithana, N.D., Ravishan, N., Sabnis, N., & Elvitigala, D.S. (2026).

VibeCoP: Enhancing the Proprioception of Center of Pressure using Vibrotactile Feedback. *Under review at ACM UIST 2026.*

Luo, H., **Balasooriya, P.**, Pathirana, R.O., Wang, H., Elvitigala, D.S., & Mueller, F. (2026). Listen to the Plant: Exploring the Design of Plant Care as an Auditory Interface to Promote Caregiving Behavior. *Conditionally accepted at ACM DIS 2026 (Provocation / Work-in-Progress).*

Luo, H., Perera, K., Montoya, M.F., Pathirana, R.O., **Balasooriya, P.**, Patibanda, R., Elvitigala, D.S., Touns Dugas, P.O., & Mueller, F. (2026). SymbioSip: Human-Plant Symbiosis through a Bidirectional Hydration System. *Accepted at ACM DIS 2026 (Interactivity).*

SKILLS SUMMARY

Programming Languages: Python, C++, C, Matlab

Tools:

- Embedded Development:** STM32Cube, PlatformIO, FreeRTOS, ESP-IDF
- PCB Design:** Altium Designer
- CAD:** SolidWorks
- Robotics & Simulation:** ROS2 Humble, Gazebo, Webots, RViz, ArduPilot
- Operating Systems & Shells:** Linux (Ubuntu), WSL, Bash
- Design and Simulation:** LTSpice, Proteus, Simulink

AI & ML Frameworks: TensorFlow, PyTorch, OpenCV, YOLO, Scikit-Learn, Edge Impulse

Web & App Dev: React, Flask, Node-RED, REST APIs, MQTT, WebSockets

Languages: English (professional proficiency), Sinhala (native proficiency)

Areas of Interest: Embedded Systems, Industrial Automation, IoT Systems, Power Electronics, Robotics, Edge AI, Human-Computer Interaction.

FootSense – Diabetic Foot Health Monitoring Platform

Dec 2025 – Present

- Led full hardware R&D including electronic design, sensor integration, and PCB development for a home-based diabetic foot ulcer early detection system.
- Developed STM32 firmware for sensor data acquisition, an Mobile companion app, cloud backend, and a clinician-facing web dashboard with thermal heatmap visualization.

VibeCoP – Vibrotactile Center of Pressure Feedback System

Jan 2025 – March 2026

- Designed ESP32-S3 dual-core FreeRTOS firmware, ESP-NOW communication, and real-time motor control.
- Co-authored research paper submitted to *ACM UIST 2026* (currently under review).

RadarFoot – mmWave Radar-Based Ground Sensing

Dec 2025 – March 2025

- Implemented gait detection and staircase detection algorithms using radar signal processing.
- Redesigned the PCB for a non-contact mmWave radar-based foot interaction sensing system.

PROJECTS

Industrial Omnidirectional Robot Platform [🔗](#)

March 2025 - July 2025

- Designed a closed-loop DC motor controller with velocity control using STM32F446RE Microcontroller.
- A custom PCB in Altium Designer and an enclosure in SolidWorks for the motor controller.
- A FreeRTOS-based firmware for real-time control, sensor data handling and communication.

Tools & technologies: STM32Cube, Embedded C, Altium Designer, SolidWorks, FreeRTOS, RS485

Smart Medibox [🔗](#)

Apr 2025 – Jun 2025

- An IoT-based automation device with time-based alarms, environmental monitoring, and remote dashboard.
- Implemented an ESP32-based system with NTP time sync, MQTT communication, OLED display UI, and Node-RED web dashboard.

Tools & technologies: ESP32, C++, MQTT, Node-RED, DHT11, NTP.

Autonomous Driving System – Formula Student (Falcon-E UOM)

Jan 2026 – Present

- Developed a full ROS 2 Humble autonomy stack for a Formula Student electric vehicle in simulation (EUFS Gazebo).
- Implemented a cone-based path planner (color separation, left-right pairing, centerline generation) and a Pure Pursuit steering controller with tunable lookahead.
- Designed a modular node pipeline: perception bridge → cone fusion → path planning → control, communicating via custom ROS 2 messages.

Tools & technologies: ROS 2 Humble, Gazebo, C++, Python, Ackermann Steering, RViz

Autonomous Drone System [🔗](#)

Sep 2025 – Nov 2025

- Developed a fully onboard autonomous drone capable of completing multi-stop precision landings on alphabetically marked pads under real outdoor conditions.
- Implemented a custom-trained YOLO-based vision system for real-time letter detection using Raspberry Pi.
- Designed autonomy logic for takeoff, waypoint flight, pad detection, touch-and-go landings, and final precision landing—without GPS.

Tools & technologies: YOLO (Custom Training), Raspberry Pi (Python, OpenCV), ArduPilot, Pixhawk.

Micromouse Robot [🔗](#)

Aug 2024 – Oct 2025

- Iterative development of a micromouse robot for algorithm optimization and performance enhancement.
- Exploration, flood-fill path planning, velocity profiling logic in STM32 C firmware.
- A custom PCB for power management and sensor interfacing.

Tools & technologies: STM32F411CEU6 (C, STM32CubeIDE), Altium Designer, FloodFill Algorithm, Sharp IR, VL53L0X ToF Sensors, UART

STM32/Vision-based Robot (SLRC 2025) [🔗](#)

Jan 2025 – Apr 2025

- Low-level STM32F446RE C firmware with velocity profiles, motor control, and sensor handling.
- OpenCV-based vision modules in Python for detecting Colors and objects on Raspberry Pi.
- Custom C firmware for all the embedded peripherals to interface with STM32F446RE.

Tools & technologies: STM32F446RE (C, STM32CubeIDE), Raspberry Pi (Python, OpenCV), UART, I2C.

Vision-Based Kobuki Robots and Simulation

Jan 2025 – Apr 2025

- Webots simulation with flood-fill, Trémaux, and DFS algorithms for maze-solving.
- OpenCV-based vision pipeline for real-world detection and navigation for Kobuki robots.

Tools & technologies: Webots, OpenCV, Python, Raspberry Pi, Ubuntu

Multi-Task Autonomous Robot

July 2024 – Dec 2024

- An robot capable of line following, maze solving, box manipulation, color detection, and timed interactions.
- C++ firmware for autonomous navigation, line following, maze solving, box manipulation and color detection.

Tools & Technologies: PlatformIO, ATmega2560, Arduino, SolidWorks, VL53L0X, TCS3200.

AWARDS AND SCHOLARSHIPS

1st Runner Up Sri Lanka Robotics Challenge  <i>A STM32/Vision Based autonomous robot to complete the given task</i>	April 2025
1st Runner Up Robofest 2025 — Open Category  <i>Vision based Autonomous Drone Navigation</i>	Oct 2025
2nd Runner Up IESL Robogames 2024  <i>Autonomous navigation for Kobuki robots using OpenCV, Webots Simulation, and Raspberry Pi</i>	April 2025
2nd Runner Up GENESIZ'25 KDU, Sri Lanka  <i>Open theme idea pitching Challenge: "Omnidirectional Robots For industries"</i>	Aug 2025
1st Place EN2533: Robot Design and Competition  <i>A Robot with autonomous navigation to complete the given task</i>	Dec 2024
Finalists Robofest 2024  <i>A micromouse robot to complete a 16 x 16 maze</i>	Oct 2024
Batch Top Honors - Information Technology - ESOFT Metro Campus, Sri Lanka <i>Awarded to the best performer out of 200+ students</i>	Dec 2019

CERTIFICATES AND COURSES

MIT16.485 - Visual Navigation for Autonomous Vehicles (Course) <i>MIT OpenCourseWare</i> • SLAM, visual odometry, perception pipelines for autonomous navigation, ROS2	2025
Advanced Kalman Filtering and Sensor Fusion <i>Udemy</i> • State estimation, Extended Kalman Filter, IMU and GPS fusion	2025
Embedded Machine Learning for Edge Computing  <i>University of Moratuwa</i> • On-device ML model development and optimization for microcontrollers	2025
Microcontroller Embedded C Programming <i>Udemy</i> • Low-level programming for ARM Cortex-M, peripheral interfacing	2025

VOLUNTEERING & LEADERSHIP EXPERIENCE

Co-Chair: IEEE Robotics Day 2025  <i>IEEE Student Branch of University of Moratuwa</i>	Mar 2025 – Apr 2025
Finance and Industrial Relations Committee Lead  <i>IEEE Student Branch of University of Moratuwa</i>	Jan 2025 – Dec 2025
Industrial Relations Committee Member <i>Electronic Club of University of Moratuwa</i>	Jan 2024 – Dec 2024

REFERENCES

Dr. Don Samitha Elvitigala

Senior Lecturer / Assistant Professor, FIT
Deputy Course Director, Monash Online (MCS)
Department of Human Centred Computing
Faculty of Information Technology
Monash University, Australia
✉ don.elvitigala@monash.edu
☎ +61-490-830 383

Dr. Peshala Jayasekara

Senior Lecturer
Department of Electronic & Telecom. Engineering
University of Moratuwa, Sri Lanka
✉ peshala@uom.lk